



Amrita School of Engineering, Chennai

Course Delivery Plan

Course Code: Course Name	23CCE215 Machine Learning	Program	CCE
L - T - P - C	3-0-0-3	Semester / Year	IV/ II
Name(s) of the faculty	Dr Gayathri M	Pre-requisite	Foundations of Data Science
Course Mentor	Dr Gayathri M	Academic Year	2025 -2026
Course Overview	This course focuses on mathematical methods behind the data interpretation, analysis, and to design/create machine learning algorithms addressing State-of-the-art approaches.		

Course Objectives		Course Outcomes	
1	To provide the foundations of machine learning	CO1	Understand the mathematical foundations of machine learning
2	To introduce supervised and unsupervised learning techniques	CO2	Understand supervised and unsupervised learning techniques
3	To enable the appreciation of machine learning techniques	CO3	Apply machine learning techniques to standard datasets
		CO4	Analyze the performance of machine learning models

Syllabus

Unit 1:

Optimization: Review of partial derivatives, gradient based optimization, Hessian and Jacobian matrices, constrained optimization – challenges, gradient descent algorithm, algorithms with adaptive learning rates.

Introduction to Machine Learning: Machine Learning Pipeline - Data Preprocessing: Standardization, Normalization, Missing data problem, Data imbalance problem, Data visualization - Setting up training, development and test sets, Cross validation, Problem of Overfitting, Bias vs Variance, Evaluation measures

Unit 2

Supervised Learning: Linear regression – single and multi-variable cases, regularization, bias and variance, Logistic regression, Classification: K-Nearest Neighbor, Naïve Bayes, Decision Tree, Random Forest, Support Vector Machine, Case study on advanced supervised ML techniques for regression and classification

Unit 3

Unsupervised learning: Clustering - K-means, DBSCAN, Gaussian Mixture Model, Parameter Estimation: MLE and Bayesian Estimate, Expectation Maximization, Case study on advanced unsupervised ML techniques for regression and classification. Artificial Neural Networks: Multi-layer Perceptron, Back Propagation Algorithm, ANN applications to classification and regression

Textbook(s)

T1. Goodfellow, I., Bengio, Y. and Courville, A., Deep learning, MIT Press, 2016.

T2. Christopher M Bishop. Pattern Recognition and Machine Learning, Springer 2010.

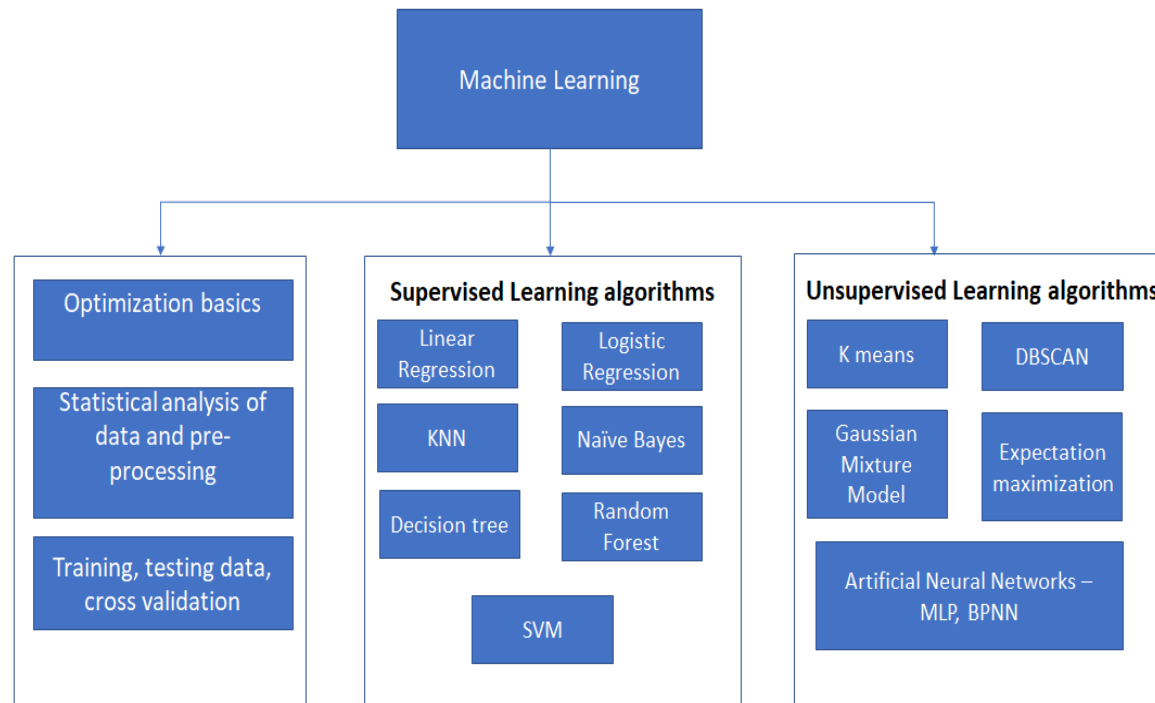
T3. Richard O. Duda, Peter E. Hart, David G. Stork. Pattern Classification. Wiley, Second Edition, 2007

Reference(s)

R1. Jiawei Han, Micheline Kamber, Jian Pei, “Data Mining: Concepts and Techniques”, 3rd Edition, Morgan Kaufmann Publishers (Elsevier), 2011.

R2. Kevin P. Murphey. Machine Learning, a probabilistic perspective. The MIT Press Cambridge, Massachusetts, 2012.

Concept Map



Evaluation and Grading

Internal (60)				External (40)	Total
Components		Marks	Total Marks	End Semester = 40	Internal + External = 100
Periodicals	Mid Term	30	30		
Continuous Assessments	VAP	10	30		
	Quiz (2)	15			
	Class participation	5			

Programme Outcome (PO)

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identity, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	To design and develop algorithms for Communication Systems
PSO2	To design and develop Embedded and Computing Systems
PSO3	To design, prototype and analyse the performance of Modern Communication Systems

CO – PO Affinity Map

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO															
CO 1	3	2										3			
CO 2	3	2										2			
CO 3	3	3										2			
CO 4	3	3										2			

3 – Strong, 2 Moderate, 1 -weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	Level 2	Level 3
55%	57%	50%	60%	70%

Class	Topics to be covered	Mode of Teaching	In-Class Activities	Out- Class Activities (E – Ref.)	CO Mapping	Reference
Unit 1						
1(L)	Review of Multi-variable Calculus	Smartboard Presentation	Discussion	https://www.youtube.com/watch?v=4QhZT agNq9Q&list=PLBh2i93oe2qv4G2Ayarkb R3OKBml0hXEg	CO1	R3
2(L)	Partial derivatives, gradient	Smartboard Presentation	Discussion	https://youtu.be/AXH9Xm6Rbfc	CO1	R3
3(L)	Hessian and Jacobian	Smartboard Presentation	Derivation and solve problems - Simulation based learning	https://youtu.be/hoYGmGxAbPY	CO1	R3
4(L)	Unconstrained Optimization – local and global minima	Smartboard Presentation	Derivation and solve problems - Simulation based learning g	https://youtu.be/pfAf7IgD7hQ?list=PL3377 D4EECF5AF70	CO1	R3
5-6(L)	Gradient descent, step-size, adaptive learning rate	Smartboard Presentation	Discussion	https://youtu.be/sDv4f4s2SB8	CO1	R3
7(L)	Introduction of machine learning, History and Different types ML applications	PPT presentation and Smartboard Teaching	Overview and industry applications of ML	https://www.youtube.com/watch?v=T3PsRW6wZSY&feature=emb_imp_woyt	CO 1	T2 & R1
8(L)	Machine Learning Pipeline - Data Preprocessing: Standardization	PPT presentation and Smartboard Teaching	Discuss about ML development	https://www.youtube.com/watch?v=2ldhbc GWI0I	CO 1	T2 & R1
9(L)	Normalization, Missing data problem	PPT presentation and Smartboard Teaching	Numerical based learning	http://www.csun.edu/~twang/595DM/Slides/Week2.pdf	CO 1	T2 & R1
10(L)	Data imbalance problem,	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=pDw_J HHvj-0	CO 1	T2 & R1

11(L)	Measuring the central tendency, Measuring the dispersion of data	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=XaHF_NhHfXwQ https://www.youtube.com/watch?v=tyfrT0cSZKo	CO 1	T2 & R1
12(L)	Data visualization	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.oreilly.com/library/view/statistics-in-a/9781449361129/ch04.html	CO 1	T2 & R1
13(L)	Setting up training, development and test sets, Cross validation	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=UJ3W_GuJ1zao	CO 1	T2 & R1
14-15(L)	Problem of Overfitting, Bias vs Variance, Evaluation measures	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=jiQamxz2ZcQ	CO 1	T2 & R1
16-17(L)	Optimization: Review of partial derivatives, gradient based optimization, Hessian and Jacobian matrices,	PPT presentation and Discussion	Group Discussion	https://www.youtube.com/watch?v=fqq_U_R4zhfI	CO 1	T 1
18(L)	Constrained optimization – challenges, gradient descent algorithm, algorithms with adaptive learning rates	PPT presentation and Discussion	Group Discussion	https://www.youtube.com/watch?v=FKCV76N9Ys0	CO 1	T 1

Unit 2

19(L)	Supervised Learning: Linear regression – single and multi-variable cases, regularization, bias and variance	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=nk2CQITm_eo	CO2, CO3, CO4	T2 & R1
20(L)	Logistic regression	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=zAULhNrnuL4	CO2, CO3, CO4	T2 & R1
21-22 (L)	Classification: K-Nearest Neighbor	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=4HKqjENq9OU https://www.youtube.com/watch?v=PNglugooJUQ	CO2, CO3, CO4	T2 & R1

Quiz 1

23(L)	Naïve Bayes	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=Y0li8ou3MhQ	CO2, CO3, CO4	T2 & R1
24-25(L)	Decision Tree	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=SRVswRH5Q7E	CO2, CO3, CO4	T2 & R1
26-27(L)	Random Forest	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=eM4uJ6XGnSM	CO2, CO3, CO4	T2 & R1
28-29(L)	Support Vector Machine	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=TtKF996oEI8	CO2, CO3, CO4	T2 & R1
30-32(L)	Case study on advanced supervised ML techniques for regression and classification	PPT presentation and Discussion	Student Presentation	https://www.youtube.com/watch?v=ocse1X_rtSI	CO2, CO3, CO4	T2 & R1
Unit 3						
33-34(L)	Unsupervised learning: Clustering - K-means	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=sU8YC4urd-w	CO2, CO3, CO4	T2 & R1
35-36(L)	DBSCAN	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=AxARUMZh0sk	CO2, CO3, CO4	T2 & R1
37(L)	Gaussian Mixture Model	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=3k0klBbFUKw	CO2, CO3, CO4	T2 & R1
38(L)	Parameter Estimation: MLE and Bayesian Estimate,	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=c42S_iHYcbY	CO2, CO3, CO4	T2 & R1

39(L)	Expectation Maximization	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=rVfZHWTwXSA&t=1118s	CO2, CO3, CO4	T2 & R1
Quiz 2						
40-41(L)	Case study on advanced unsupervised ML techniques for regression and classification	PPT presentation and Discussion	Student presentation	https://www.youtube.com/watch?v=z18nw4adsx4	CO2, CO3, CO4	T2 & R1
42-43(L)	Artificial Neural Networks: Multi-layer Perceptron,	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=xbYgKoG4x2g&list=PL53BE265CE4A6C056	CO2, CO3, CO4	T2 & R1
44-45(L)	Back Propagation Algorithm	PPT presentation and Smartboard Teaching	Numerical based learning	https://www.youtube.com/watch?v=y0wNuFFPGuI	CO2, CO3, CO4	T2 & R1
46-47(L)	ANN applications to classification and regression	PPT presentation and Discussion	Student presentation	https://www.youtube.com/watch?v=Wo5dMEP_BbI&list=PLQVvvaa0QuDcjD5BAw2DxE6OF2tius3V3	CO2, CO3, CO4	T2 & R1

Attendance Requirement:

- (i) All students must maintain a minimum of 75% attendance in each course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.
- (ii) Students who fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams. Requests for exceptions will not be considered under any circumstances.
- (iii) Attendance will be calculated up to three days prior to the commencement of the mid-term and end-semester exams, as per the academic calendar.

Faculty

Course Mentor

Course Moderator

Chairperson/ECE